

Alireza Farshin | Curriculum Vitae

 [aliireza.github.io](https://github.com/aliireza) •
  Alireza Farshin •
  [aliireza](https://www.youtube.com/aliireza)
 [alireza-farshin](https://www.linkedin.com/in/alireza-farshin) •
  [alirezafarshin](https://twitter.com/alirezafarshin)

I am a distributed systems researcher at NVIDIA. My research interests include computer networks and networked systems. During my doctoral studies, I improved the performance of Network Functions Virtualization (NFV) service chains by using **low-level optimization** techniques.  [Watch](#)

Work Experience

- Networking Software & Systems Research Team at NVIDIA** **Stockholm, Sweden**
Distributed Systems Researcher/Senior Software Architect *January 2024–now*
 - Next-generation programmable NICs.
 - Network simulation for AI/ML workloads (see [MLSynth](#) and [CheckWork](#)).
 - Code generation using LLMs.
- Connected Intelligence Unit at RISE Research Institutes of Sweden** **Stockholm, Sweden**
Senior Researcher in AI / Machine Learning and Networking *August 2023–January 2024*
 - Improving packet processing at multi-100-Gbps rates (see [FAJITA](#)).
 - Using large language models (LLMs) to build and configure networked systems (see [NetBuddy](#), [FlowMage](#), and [NetConfEval](#)).
 - Developing pruning techniques and improving inference of LLMs (see [W4]).
- Network Systems Laboratory (NSLab) at KTH** **Stockholm, Sweden**
Postdoctoral Researcher *March 2023–August 2023*
- Network Systems Laboratory (NSLab) at KTH** **Stockholm, Sweden**
Doctoral Researcher *August 2017–March 2023*
- ICT Doctoral Programme Council at KTH** **Stockholm, Sweden**
Student Representative of the Division of Communication Systems ([CoS](#)) *May 2018–December 2020*
- Mobile Telecommunication Company of Iran (MCCI)** **Tehran, Iran**
Portal Specialist *December 2015–June 2016*
 Vendor Manager & Portal/Application Supervisor:
 - eCare Application: My MCI Application for [iOS](#) and [Android](#)
 - eSales Website: [eVoucher](#)
- CafeYab** **Tehran, Iran**
Co-founder and CEO *Fall-2013*
 An application for iOS and [Android](#) for finding nearby Coffee Shops
- Informatics Services Corporation (ISC)** **Tehran, Iran**
Internship *June 2013–September 2013*
 - Ported an RF unit controller from PIC-16F877A to AtMega64A and tested the new module.
 - Designed a remote-control system with HM-T and HM-R FSK modules.

Publications

Conference Publications

- [C1] Hamid Ghasemirahni, **Alireza Farshin**, Mariano Scazzariello, Gerald Q. Maguire Jr., Dejan Kostić, Marco Chiesa. FAJITA: Stateful Packet Processing at 100 Million pps, *The 20th International Conference on emerging Networking EXperiments and Technologies (CoNEXT)*. 2024. [Download](#)

- [C2] Changjie Wang, Mariano Scazzariello, **Alireza Farshin**, Simone Ferlin, Dejan Kostić, Marco Chiesa. NetConfEval: Can LLMs Facilitate Network Configuration?, *The 20th International Conference on emerging Networking EXperiments and Technologies (CoNEXT)*. 2024. [IRTF/IETF ANRP Winner!] [Download](#)
- [C3] Hamid Ghasemirahni, Tom Barbette, Georgios Katsikas, **Alireza Farshin**, Massimo Gironi, Amir Roozbeh, Marco Chiesa, Gerald Q. Maguire Jr., Dejan Kostić. Packet Order Matters! Improving Application Performance by Deliberately Delaying Packets In *19th USENIX Symposium on Networked Systems Design and Implementation (NSDI)*. 2022. Acceptance rate (Spring): 28/104 $\approx$ 26.9%. [Community Award Winner!] [Download](#)
- [C4] **Alireza Farshin**, Tom Barbette, Amir Roozbeh, Gerald Q. Maguire Jr., Dejan Kostić. PacketMill: Toward per-core 100-Gbps Networking In *International Conference on Architectural Support for Programming Languages and Operating Systems (ASPLOS)*. 2021. Acceptance rate: 75/398 $\approx$ 18.8%. [Download](#)
- [C5] **Alireza Farshin**, Amir Roozbeh, Gerald Q. Maguire Jr., Dejan Kostić. Reexamining Direct Cache Access to Optimize I/O Intensive Applications for Multi-hundred-gigabit Networks In *USENIX Annual Technical Conference (ATC)*. 2020. Acceptance rate: 65/348 $\approx$ 18.6%. [Download](#)
- [C6] **Alireza Farshin**, Amir Roozbeh, Gerald Q. Maguire Jr., Dejan Kostić. Make the Most out of Last Level Cache in Intel Processors In *The European Conference on Computer Systems (EuroSys)*. 2019. Acceptance rate: 45/207 $\approx$ 21.7%. [Download](#)

Journal Publications.....

- [J1] **Alireza Farshin**, Luigi Rizzo, Khaled Elmeleegy, Dejan Kostić. Overcoming the IOTLB wall for multi-100-Gbps Linux-based networking In *PeerJ Computer Science (PeerJ CS)*. 2023. **Impact factor: 2.41**. [Download](#)
- [J2] **Alireza Farshin**, Saeed Sharifian. A modified knowledge-based ant colony algorithm for virtual machine placement and simultaneous routing of NFV in distributed cloud architecture In *The Journal of Supercomputing (SUPE)*. 2019. **Impact factor: 2.469**. [Download](#)
- [J3] **Alireza Farshin**, Saeed Sharifian. A chaotic grey wolf controller allocator for Software Defined Mobile Network (SDMN) for 5th generation of cloud-based cellular systems (5G) In *The Journal of Computer Communications (COMCOM)*. 2017. **Impact factor: 2.816**. [Download](#)
- [J4] **Alireza Farshin**, Saeed Sharifian. MAP-SDN: a metaheuristic assignment and provisioning SDN framework for cloud datacenters In *The Journal of Supercomputing (SUPE)*. 2017. **Impact factor: 2.469**. [Download](#)

Patent Applications.....

- [P1] **Alireza Farshin**, Daniel Marcovitch, Liran Liss. Network Integration. US Patent Application 19/542,426. Filed in February 2026.
- [P2] **Alireza Farshin**, Liran Liss, Adel Sefiane. Generating Plans to Perform Workloads. US Patent Application 19/461,416. Filed in January 2026.
- [P3] **Alireza Farshin**, Liran Liss, Adel Sefiane, Ilias Marinos. Proxy Communication between Processing Environments. US Patent Application 19/369,134. Filed in October 2025.
- [P4] **Alireza Farshin**, Omri Kahalon, Liran Liss, Aviad Shaul Yehezkel, Vishwanath Venkatesan. Distributed Memory with Pre-Computed Key and Value (KV) Entries for Inference Operations. US Patent Application 19/173,029. Filed in April 2025.
- [P5] **Alireza Farshin**, Omri Kahalon, Liran Liss, Aviad Shaul Yehezkel, Vishwanath Venkatesan. Distributed Memory with Subsets of Model Weights or Parameters based on Inference Task Types. US Patent Application 19/173,027. Filed in April 2025.

- [P6] **Alireza Farshin**, Omri Kahalon, Liran Liss, Aviad Shaul Yehezkel, Vishwanath Venkatesan. Distributed Memory of Inference Operations. US Patent Application 19/173,023. Filed in April 2025.
- [P7] **Alireza Farshin**, Omri Kahalon, Vishwanath Venkatesan, Timothy Stamler. Orchestration of Distributed Inference Operations. US Patent Application 18/926,233. [Download](#)
- [P8] Amir Roozbeh, **Alireza Farshin**, Marco Chiesa, Dejan Kostić. Network Entity and Method Performed Therein for Handling one or more Packets in a Computer Environment (*Other name*: System and Methods for Minimizing Branch Mispredictions and Executing Highly Optimized Code for Networking Applications). PCT Application PCT/SE2023/050880. [Download](#)
- [P9] Amir Roozbeh, **Alireza Farshin**, Marco Chiesa. Network Entity and Method Performed Therein for Handling one or more Packets in a Computer Environment (*Other name*: System and Methods for Programmatically Storing Packet Payloads in the Per-Port Queue Memory). US Provisional Patent Application 63/511,198. Filed in June 2023.
- [P10] Amir Roozbeh, **Alireza Farshin**, Marco Chiesa. System and Method Performed Therein for Handling one or more Packets in a Computer Environment (*Other name*: System and Methods for Disaggregated Packet Construction). PCT Application PCT/SE2023/050538. [Download](#)
- [P11] Amir Roozbeh, **Alireza Farshin**, Marco Chiesa. Entity and Method Performed Therein for Handling Packets in a Computer Environment (*Other name*: System and Methods for Performing Millions Low-Latency Key-Value Insertion on Switches). PCT Application PCT/SE2023/051174. [Download](#)
- [P12] Amir Roozbeh, **Alireza Farshin**, Marco Chiesa, Dejan Kostić, Hamid Ghasemirahni. Hint Entity, Receiver Node, System and Methods Performed Therein for Handling Data in a Computer Environment (*Other name*: System and Methods for Network-Accelerated State Prefetching). PCT Application PCT/SE2022/051036. [Download](#)
- [P13] Amir Roozbeh, Chakri Padala, **Alireza Farshin**, Dejan Kostić, Gerald Q. Maguire Jr. Processing Unit, Packet Handling Unit, Arrangement and Methods for Handling Packets (*Other name*: System and Methods for Probing and Polling Multiple I/O Operations on I/O Devcies for Priority-Based Packet Processing). PCT Application PCT/SE2022/050710. [Download](#)
- [P14] Amir Roozbeh, **Alireza Farshin**, Marco Chiesa, Tom Barbette, Dejan Kostić. Packet Processing Including an Ingress Packet Part Distributor. PCT Application PCT/EP2023/063619. [Download](#)
- [P15] Amir Roozbeh, **Alireza Farshin**, Dejan Kostić. System and Method for Organizing Physical Queues into Virtual Queues. PCT Application PCT/EP2022/051103. [Download](#)
- [P16] Amir Roozbeh, **Alireza Farshin**, Marco Chiesa, Fabio Luciano Verdi. System and Method for Accurate Traffic Monitoring on Multi-Pipeline Switches. PCT Application PCT/EP2021/084572. [Download](#)
- [P17] Amir Roozbeh, Chakri Padala, **Alireza Farshin**. System and Method for Cache pooling and Efficient Usage and I/O Transfer in disaggregated and Multi-Processor Architectures via Processor Interconnect. PCT Application PCT/SE2021/051016. [Download](#) ([US12499046B2](#) [Granted](#))
- [P18] Amir Roozbeh, **Alireza Farshin**, Chakri Padala, Dejan Kostić, Gerald Q. Maguire Jr. System, Method, and Apparatus for Fine-grained Control of I/O Data Placement in Memory Subsystem. PCT Application PCT/SE2021/050803. [Download](#)
- [P19] Amir Roozbeh, **Alireza Farshin**, Tom Barbette, Dejan Kostić, Gerald Q. Maguire Jr. Methods and Systems for Efficient Metadata and Data Delivery between a Network Interface and Applications. PCT Application PCT/IB2021/052976. [Download](#)
- [P20] Amir Roozbeh, **Alireza Farshin**, Dejan Kostić, Gerald Q. Maguire Jr. Method and System for Efficient Input/Output Transfer in Network Devices. PCT Application PCT/SE2020/051107 [Download](#) ([US12341678B2](#) [Granted](#)) & PCT/SE2020/051108 [Download](#) ([US12425351B2](#) [Granted](#)).
- [P21] Amir Roozbeh, **Alireza Farshin**, Dejan Kostić, Gerald Q. Maguire Jr, Hamid Ghasemirahni, Tom Barbette. Reordering and Reframing Packets. PCT Application PCT/IB2020/054991. [Download](#)

- [P22] Chakri Padala, Amir Roozbeh, **Alireza Farshin**, Dejan Kostić, Gerald Q. Maguire Jr. Efficient Loading of Code Portions to a Cache. PCT Application PCT/SE2020/050527. [Download](#)
- [P23] Amir Roozbeh, **Alireza Farshin**, Dejan Kostić, Gerald Q. Maguire Jr. Entities, System and Methods Performed Therein for Handling Memory Operations of an Application in a Computer Environment. PCT Application PCT/SE2019/050948. [Download](#) (US12111766B2 [Granted](#))
- [P24] Amir Roozbeh, **Alireza Farshin**, Dejan Kostić, Gerald Q. Maguire Jr. Methods and Devices for Controlling Memory Handling. PCT Application PCT/SE2020/050161. [Download](#) (US12111768B2 [Granted](#))
- [P25] Amir Roozbeh, Dejan Kostić, Gerald Q. Maguire Jr., **Alireza Farshin**. Memory Allocation in a Hierarchical Memory System. PCT Application PCT/SE2019/050596. [Download](#) (US12293227B2 [Granted](#))
- [P26] Amir Roozbeh, **Alireza Farshin**, Dejan Kostić, Gerald Q. Maguire Jr. Methods and Nodes for Handling Memory. PCT Application PCT/SE2018/051311. [Download](#) (US11714753B2 [Granted](#))

Workshop Papers, Extended Abstracts, Preprints, Technical Reports, Demo, and Posters....

- [W1] Eldar Hasanov, Adel Sefiane, **Alireza Farshin**, Marios Kogias. CheckWork: Enabling Trace-Driven Analysis of Checkpointing Overhead in Distributed ML Training In *The 10th Asia-Pacific Workshop on Networking (APNet)*. 2026. [Download](#)
- [W2] Francesco De Luca, Francesco De Nadai, Mariano Scazzariello, Tommaso Caiazzzi, **Alireza Farshin**, Marco Chiesa, Giuseppe Di Battista. Aggregate Local, Sync Global: A Hierarchical Approach to Efficient Geo-Distributed LLM Training In *First Workshop on Computation over Heterogeneous Networks (NetCompute)*. 2026. [Download](#)
- [W3] Adel Sefiane, **Alireza Farshin**, Marios Kogias. MLSynth: Towards Synthetic ML Traces In *The 2nd Workshop on Networks for AI Computing (NAIC)*. 2025. [Download](#)
- [W4] Laura Puccioni, **Alireza Farshin**, Mariano Scazzariello, Changjie Wang, Marco Chiesa, Dejan Kostić. Deriving Coding-Specific Sub-Models from LLMs using Resource-Efficient Pruning In *The Second International Workshop on Large Language Models for Code (LLM4Code)*. 2025. [Download](#)
- [W5] Hamid Ghasemirahni, **Alireza Farshin**, Dejan Kostić, Marco Chiesa. Just-in-Time Packet State Prefetching, *ArXiv Preprint*. 2024. [Download](#)
- [W6] Hamid Ghasemirahni, **Alireza Farshin**, Mariano Scazzariello, Marco Chiesa, Dejan Kostić. Deploying Stateful Network Functions Efficiently using Large Language Models In *The Workshop on Machine Learning and Systems (EuroMLSys)*. 2024. [Download](#)
- [W7] Changjie Wang, Mariano Scazzariello, **Alireza Farshin**, Dejan Kostić, Marco Chiesa. Making Network Configuration Human Friendly, *ArXiv Preprint*. 2023. [Download](#)
- [W8] **Alireza Farshin**, Amir Roozbeh, Christian Schulte, Gerald Q. Maguire Jr., Dejan Kostić. Scheduling - A Secret Sauce For Resource Disaggregation, *Technical Report*. 2021. [Download](#)
- [W9] **Alireza Farshin**, Tom Barbette, Amir Roozbeh, Gerald Q. Maguire Jr., Dejan Kostić. *PacketMill*: Toward per-core 100-Gbps Networking In *International Conference on Architectural Support for Programming Languages and Operating Systems (ASPLOS)*. 2021. [Download](#)
- [W10] **Alireza Farshin**, Amir Roozbeh, Gerald Q. Maguire Jr., Dejan Kostić. Optimizing Intel Data Direct I/O Technology for Multi-hundred-gigabit Networks In *The European Conference on Computer Systems (EuroSys)*. 2020. [Download](#)
- [W11] **Alireza Farshin**, Amir Roozbeh, Gerald Q. Maguire Jr., Dejan Kostić. Make the Most out of Last Level Cache in Intel Processors In *The European Conference on Computer Systems (EuroSys)*. 2019. [Download](#)

Funded Projects

-  **SAFIR - Secure AI For Intelligent Resilience and Confidentiality in the Cloud**
 ~27 Million SEK November 2025–October 2028
 Funded by Vinnova - Advanced digitalization - Industrial needs-driven innovation 2025
 Acted as representative from NVIDIA
 Other partners: KTH (PI: Marco Chiesa), RISE, CanaryBit, Ericsson, Saab AB, and RedHat AB
-  **SEMLA - Securing Enterprises via Machine-Learning-based Automation**
 ~9.5 Million SEK November 2023–October 2025
 Funded by Vinnova - Cyber Security for Industrial Advanced Digitalization 2023
 Acted as one of the co-PIs from RISE AB
 Other partners: KTH (led by Marco Chiesa), Saab AB, and RedHat AB
-  **Realizing Low-Latency Internet Services via Low-Level Optimization of NFV Service Chains**
 \$140,000 USD August 2021–August 2023
 Funded by Google PhD Fellowship in Systems and Networking

Open-Source Contributions

-  **iommu-bench**: Understanding the IOTLB Wall for Multi-100-Gbps Linux-based Networking [[Link](#)]
-  **DDC-RA**: A Constrained-based Scheduler for Disaggregated Data Centers (DDC) [[Link](#)]
-  **PacketMill**: Toward per-core 100-Gbps Networking [[Link](#)]
-  **ddio-bench**: Understanding Intel Data Direct I/O Technology [[Link](#)]
-  **Slice-aware Memory Management**: Exploiting NUCA Characteristic of LLC in Intel Processors [[Link](#)]
-  **CacheDirector**: Sending Packets to the Right Slice by Exploiting Intel Last-Level Cache Addressing [[Link](#)]

Education

-  **KTH Royal Institute of Technology** **Stockholm, Sweden**
Ph.D. in Information and Communication Technology, School of EECS August 2017–March 2023
 Advisors: Prof. [Dejan Kostić](#) and Prof. [Gerald Q. Maguire Jr.](#)
 Dissertation Title: Realizing Low-Latency Packet Processing on Multi-Hundred-Gigabit-Per-Second Commodity Hardware (see my [Dissertation](#))
 I also received my [licentiate](#) degree (Halfway to Ph.D.) in June 2019, see my [Thesis](#).
-  **Amirkabir University of Technology** **Tehran, Iran**
M.Sc. Electrical Engineering - Digital Electronic Circuits, Department of EE September 2015–July 2017
 Advisor: Associate Prof. [Saeed Sharifian](#)
 Thesis: Resource Allocation in Software-Defined Networks for 5G Applications
 I used bio-inspired metaheuristic algorithms to perform resource allocation.
-  **Sharif University of Technology** **Tehran, Iran**
B.Sc. Electrical Engineering - Electronics, EE Department September 2010–July 2015
 Advisor: Associate Prof. [Mehran Jahed](#)
 Thesis: Design of Exoskeletal System for Wrist and Forearm

Honors, Awards, and Professional Services

2025: PC Member for [eBPF'25](#).

2025: NetConfEval [C2] received IRTF/IETF Applied Networking Research Prize 2025.
2024: Packet Order Matters! [C3] was featured in the WIPO Green Technology Book.
2024: Giving a talk with Luigi Rizzo at Netdev 0x18 about the IOTLB Wall.
2024: PC Member for eBPF'24.
2023: Faculty at Digital Futures.
2023: Reviewer for IEEE Computer Architecture Letters.
2022: Packet Order Matters! [C3] was featured in the Ericsson Blog and KTH.
2022: "Framtidens Forskning" has published a Swedish article on my research.
2022: PC Member for SIGCOMM'22 posters and demos program.
2022: Packet Order Matters! [C3] received the "Community Award" at NSDI'22.
2022: Giving a talk at Cisco Engineering Switzerland about Optimization Techniques for NFV.
2021: Awarded Google PhD Fellowship 2021 in Systems and Networking. [Interview with KTH EECS]
2021: PacketMill [C4] was featured in the Ericsson Blog.
2021: Giving a talk with Tom Barbette at FOSDEM'21. [Watch]
2020: EuroSys'20 Shadow Program Committee.
2019: CacheDirector [C6] was featured in the Ericsson Blog, Tech Xplore, AlphaGalileo, and KTH.
2018: External Reviewer for NSDI'19.
2015: **Ranked 107th** among more than 20,000 participants in Iran's universities entrance exam for M.Sc.
2010: **Ranked 46th** among more than 460,000 participants in Iran's universities entrance exam for B.Sc.

Skills

Languages: English (Fluent), Persian (Native), Swedish (Novice)

Programming Languages: C/C++, Python, MATLAB, Scala, R, Assembly-X86, bash.

Tools & Libraries: DPDK, FastClick, Perf, LLVM, TensorFlow, Pandas, Spark, Gecode, Git, gnuplot, L^AT_EX.

Hobbies

Playing Piano and Bass Guitar, Jamming with Friends, Reading Books, Watching Movies and TV Series.